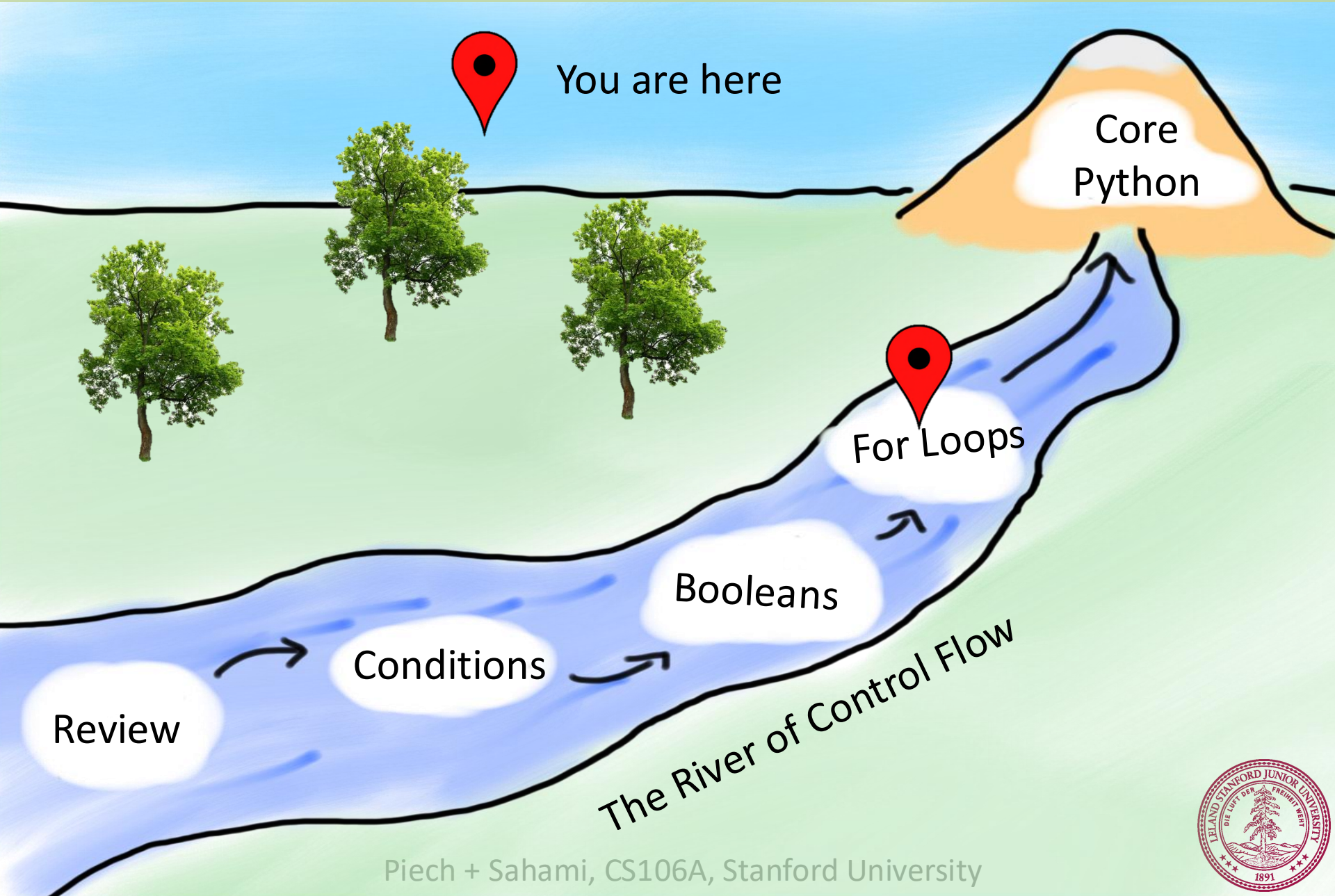


Control Flow Revisited

Chris Piech and Mehran Sahami
Stanford University

Today's Route



How would you print “Python rocks socks” 100 times

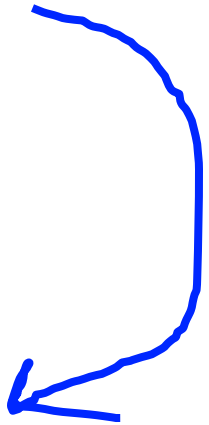
For Loop Redux

```
def main():
    for i in range(100):
        print("Python rocks socks!")
```




For Loop Redux

```
for i in range(100):
    print("Python rocks socks!")
```



```
i = 0
while i < 100:
    print("Python rocks socks!")
    i = i + 1
```

For Loop Redux

Create a counting variable i

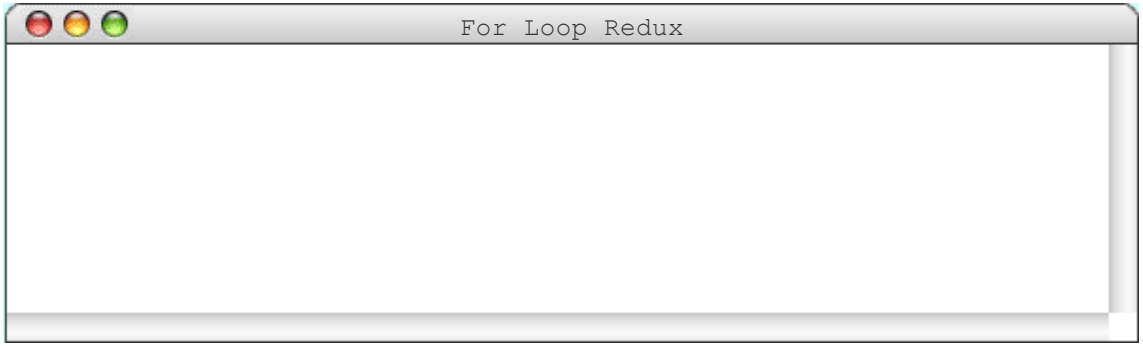
Which takes on the values 0 to 99 one at a time

```
for i in range(100):
    print("Python rocks socks!")
```

For Loop Redux

range(3) -> [0, 1, 2]

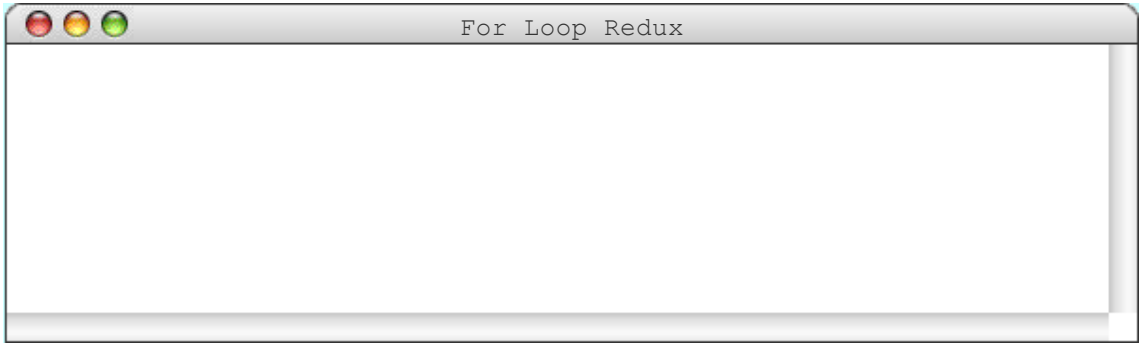
```
for i in range(3):
    print("Python rocks socks!")
```



For Loop Redux

range(3) -> [0, 1, 2]

```
for i in range(3):
    print("Python rocks socks!")
```

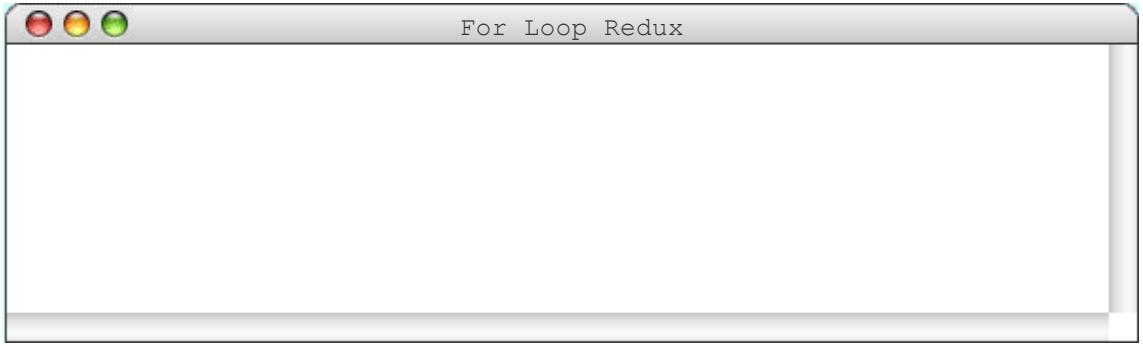


For Loop Redux

i	0

range(3) -> [0, 1, 2]

```
for i in range(3):
    print("Python rocks socks!")
```

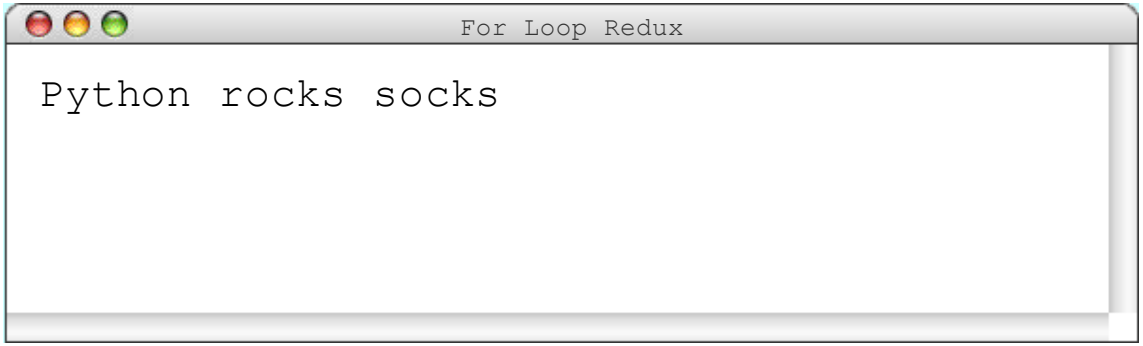


For Loop Redux

i	0

range(3) -> [0, 1, 2]

```
for i in range(3):
    print("Python rocks socks!")
```

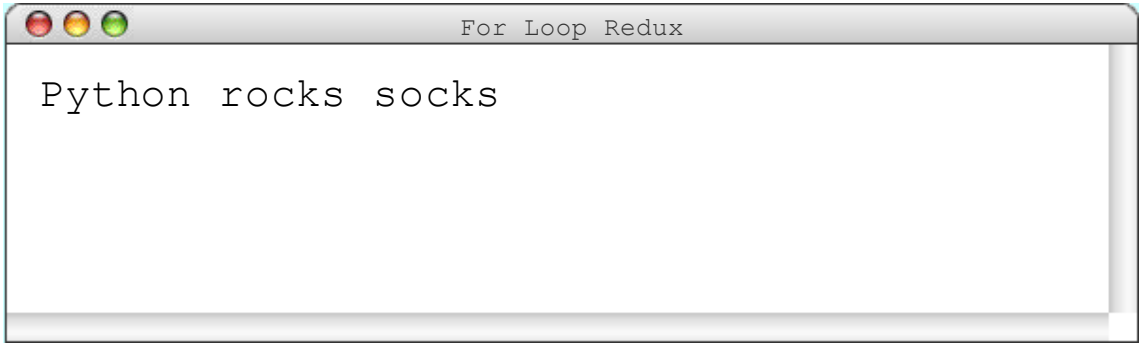


For Loop Redux

i	0

range(3) -> [0, 1, 2]

```
for i in range(3):
    print("Python rocks socks!")
```

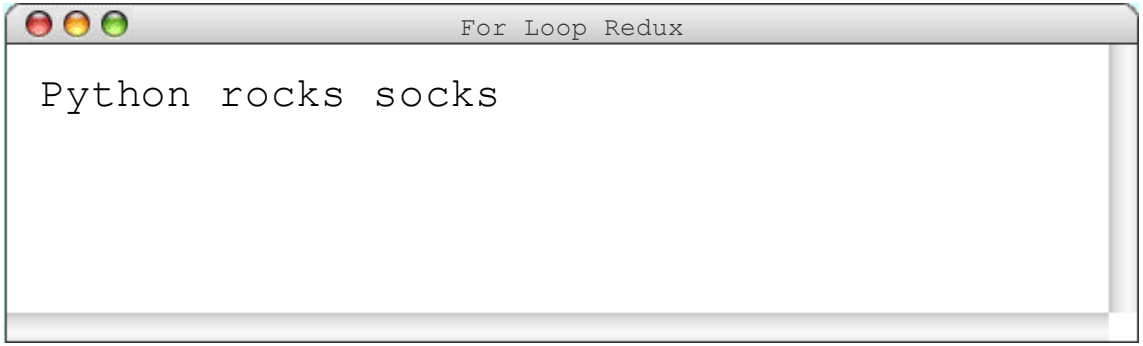


For Loop Redux

i 1

range(3) -> [0, 1, 2]

```
for i in range(3):
    print("Python rocks socks!")
```

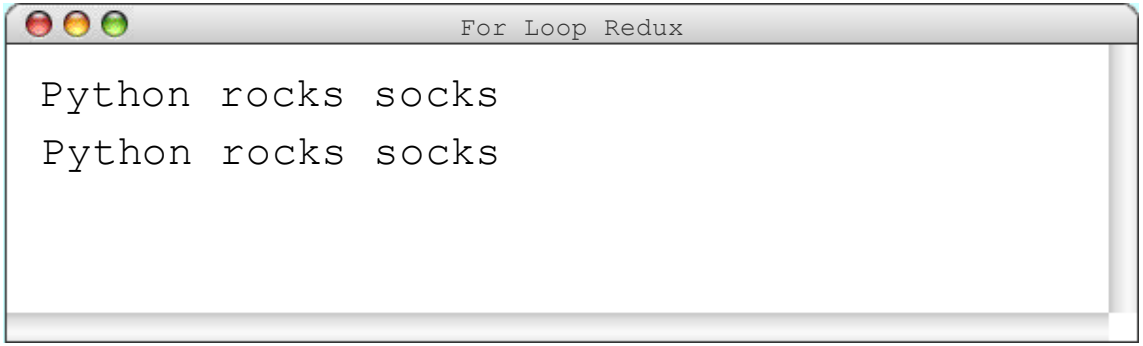


For Loop Redux

i 1

range(3) -> [0, 1, 2]

```
for i in range(3):
    print("Python rocks socks!")
```

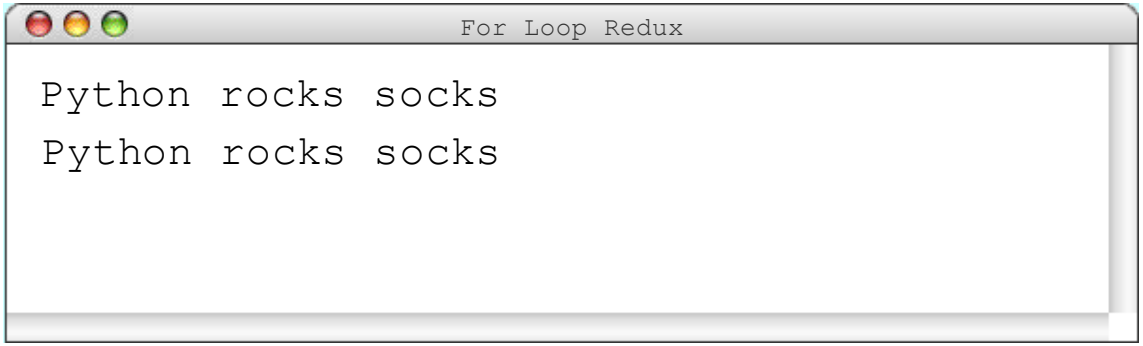


For Loop Redux

i 1

range(3) -> [0, 1, 2]

```
for i in range(3):
    print("Python rocks socks!")
```

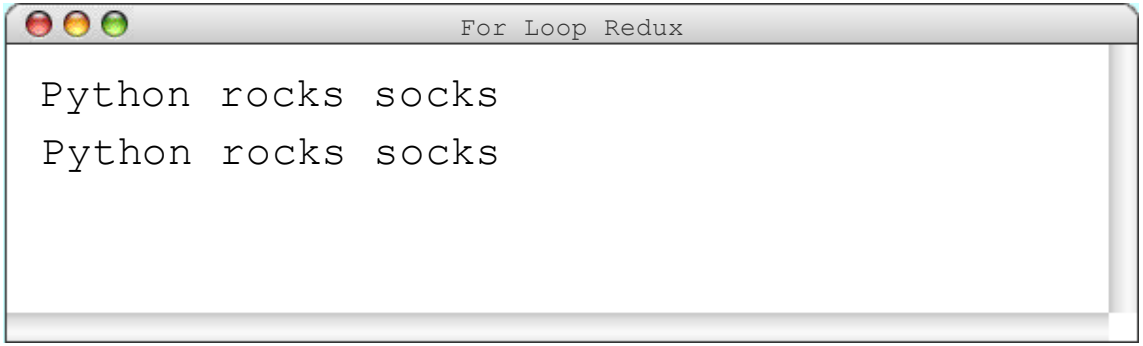


For Loop Redux

i 2

range(3) -> [0, 1, 2]

```
for i in range(3):
    print("Python rocks socks!")
```

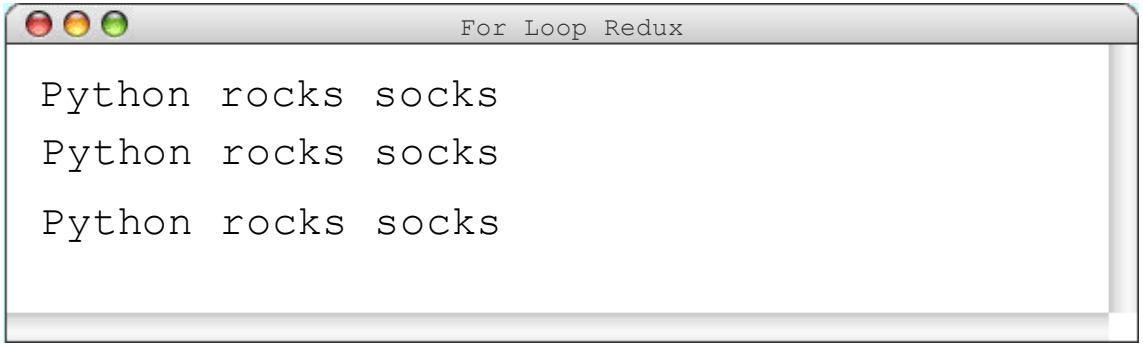


For Loop Redux

i | 2

range(3) -> [0, 1, 2]

```
for i in range(3):
    print("Python rocks socks!")
```

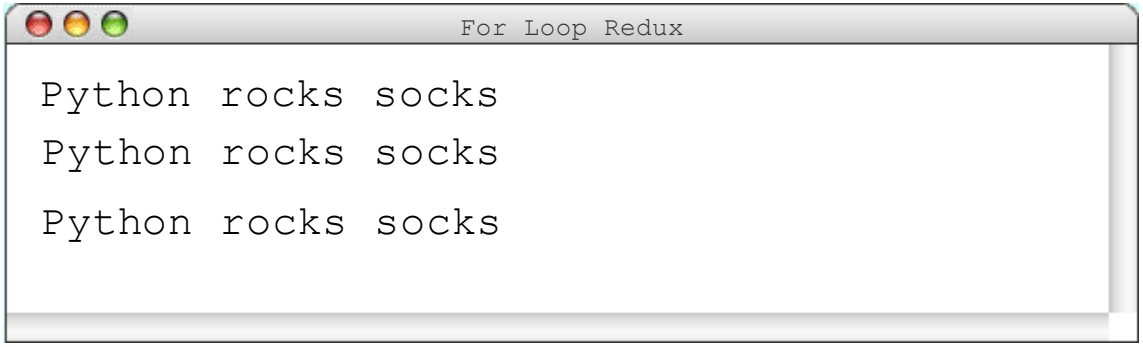


For Loop Redux

i 2

range(3) -> [0, 1, 2]

```
for i in range(3):
    print("Python rocks socks!")
```

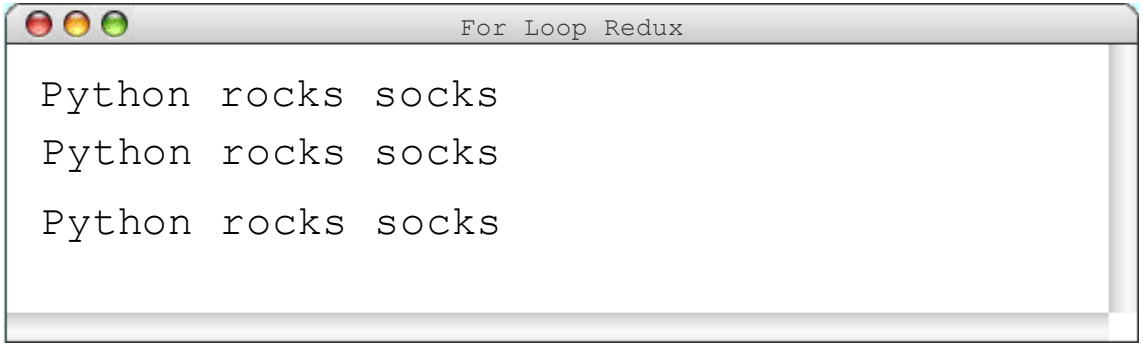


For Loop Redux

i 3

range(3) -> [0, 1, 2]

```
for i in range(3):
    print("Python rocks socks!")
```

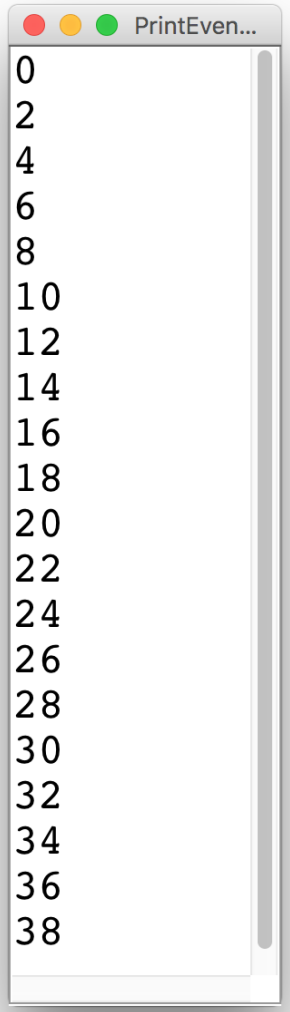


You can use the for loop variable



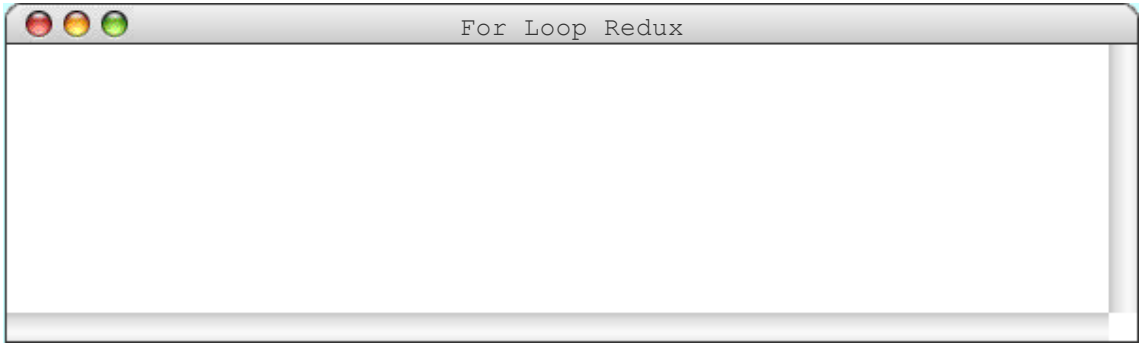
How would you print the first 100 even numbers?

Printing Even Numbers



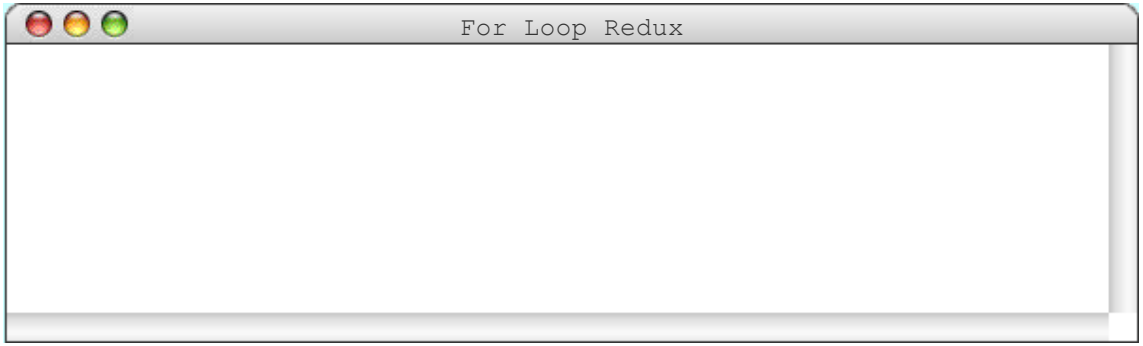
Printing Even Numbers

```
for i in range(3):
    print(i * 2)
```



Printing Even Numbers

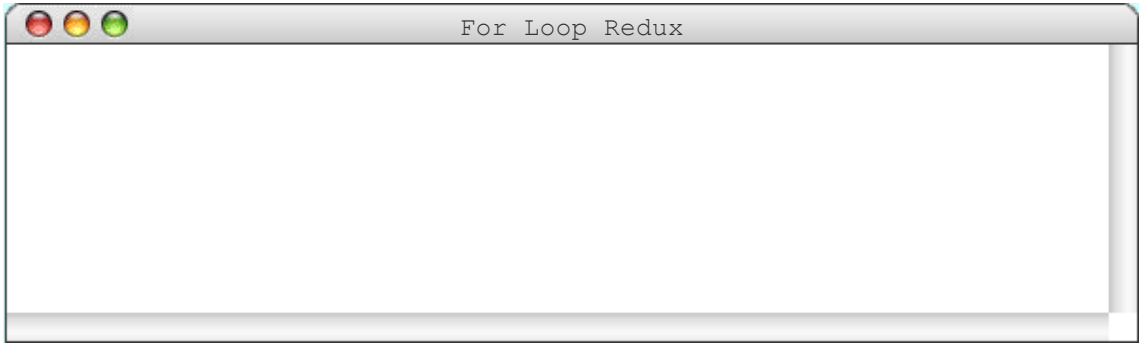
```
for i in range(3):
    print(i * 2)
```



Printing Even Numbers

i | 0

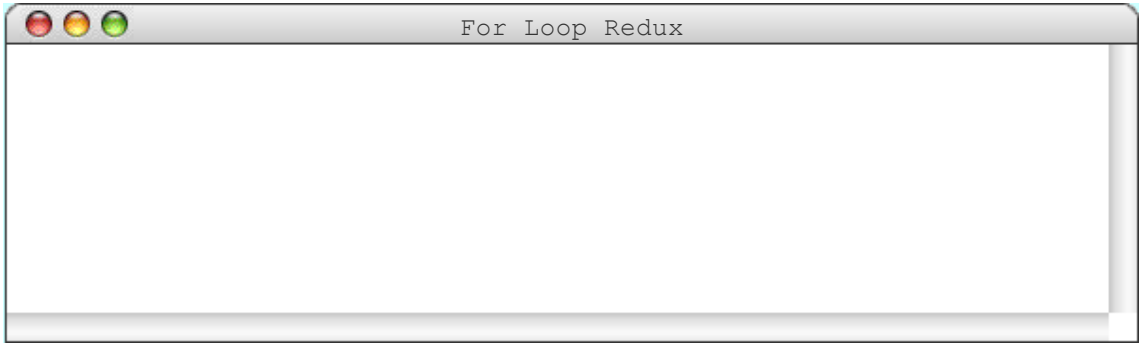
```
for i in range(3):
    print(i * 2)
```



Printing Even Numbers

i	0

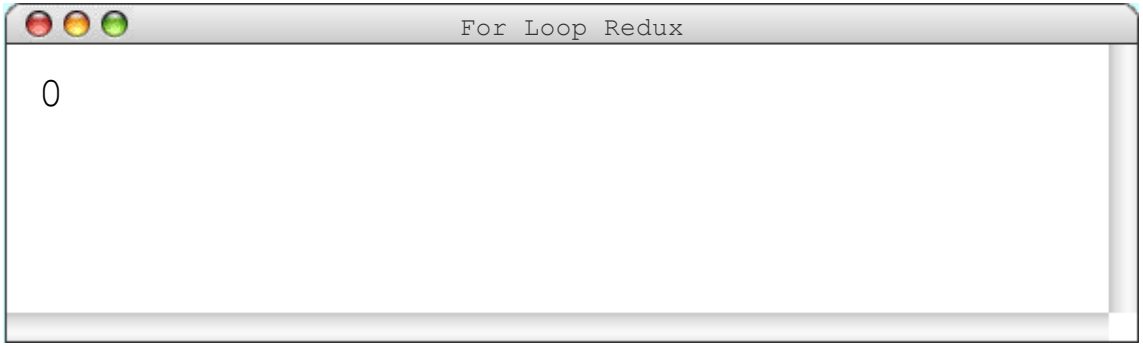
```
for i in range(3):
    print(i * 2)
```



Printing Even Numbers

i	0

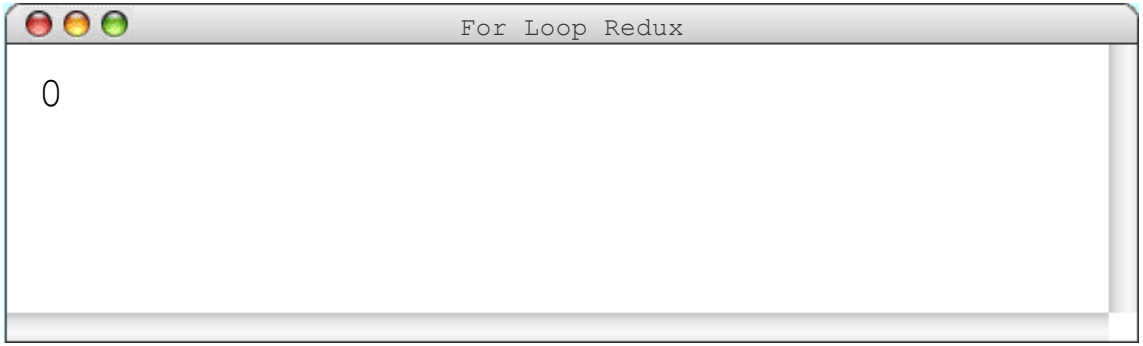
```
for i in range(3):
    print(i * 2)
```



Printing Even Numbers

i | 1

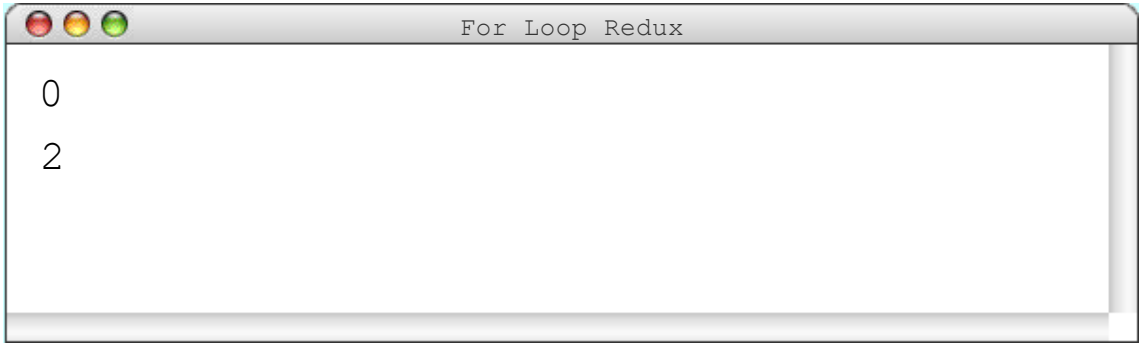
```
for i in range(3):
    print(i * 2)
```



Printing Even Numbers

i | 1

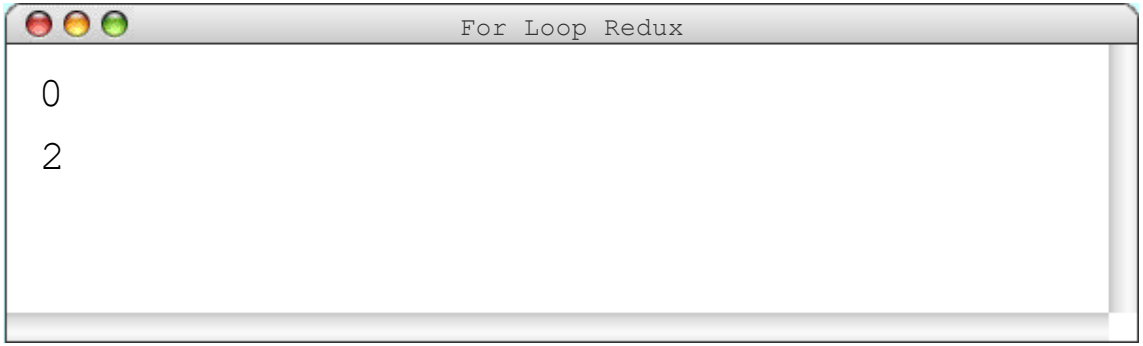
```
for i in range(3):
    print(i * 2)
```



Printing Even Numbers

i | 2

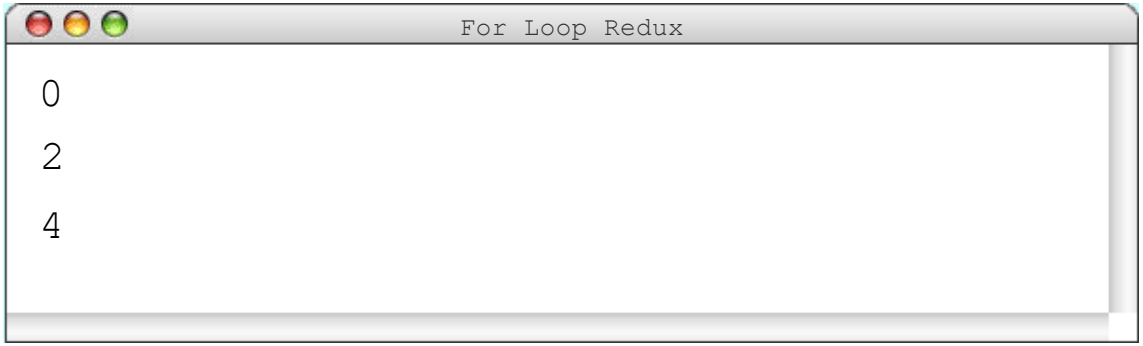
```
for i in range(3):
    print(i * 2)
```



Printing Even Numbers

i	2

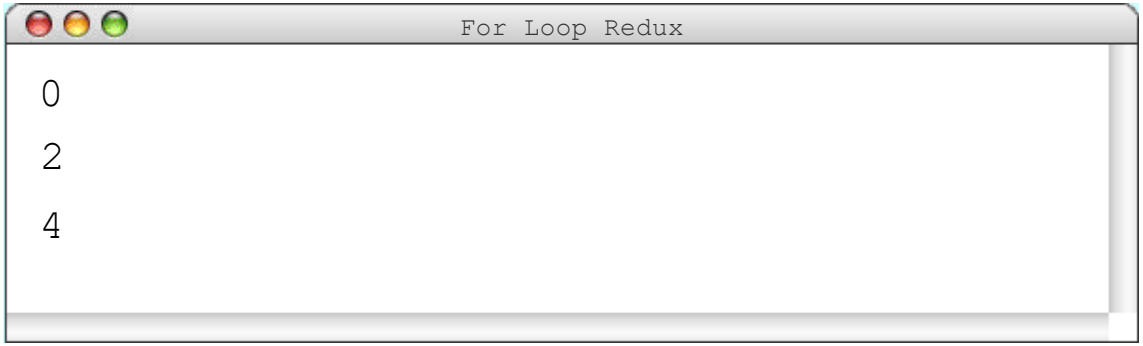
```
for i in range(3):
    print(i * 2)
```



Printing Even Numbers

i | 3

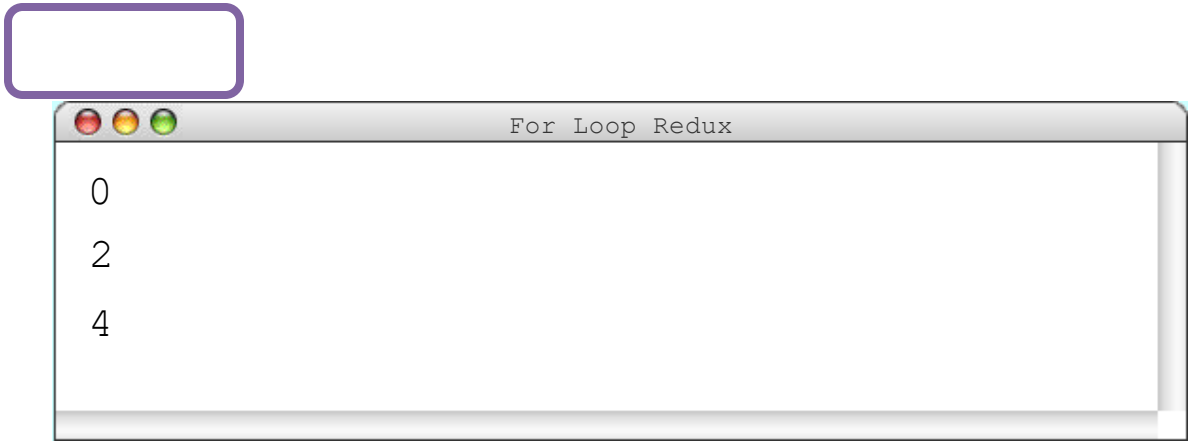
```
for i in range(3):
    print(i * 2)
```



Printing Even Numbers

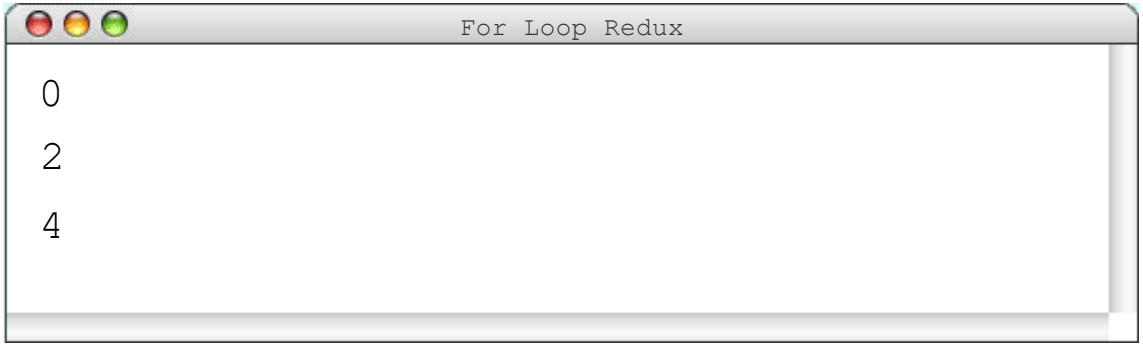
i	3

```
for i in range(3):
    print(i * 2)
```



Printing Even Numbers

```
for i in range(3):
    print(i * 2)
```



Printing Even Numbers

```
# our solution
for i in range(3):
    print(i * 2)
```

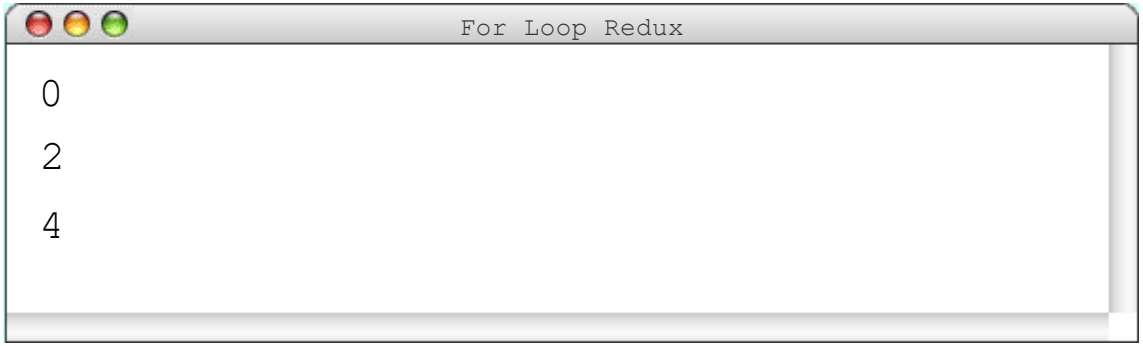
0, 1, 2

```
# equivalently
for i in range(0, 6, 2):
    print(i)
```

Start at 0

Stop before 6

Skip by 2 each time

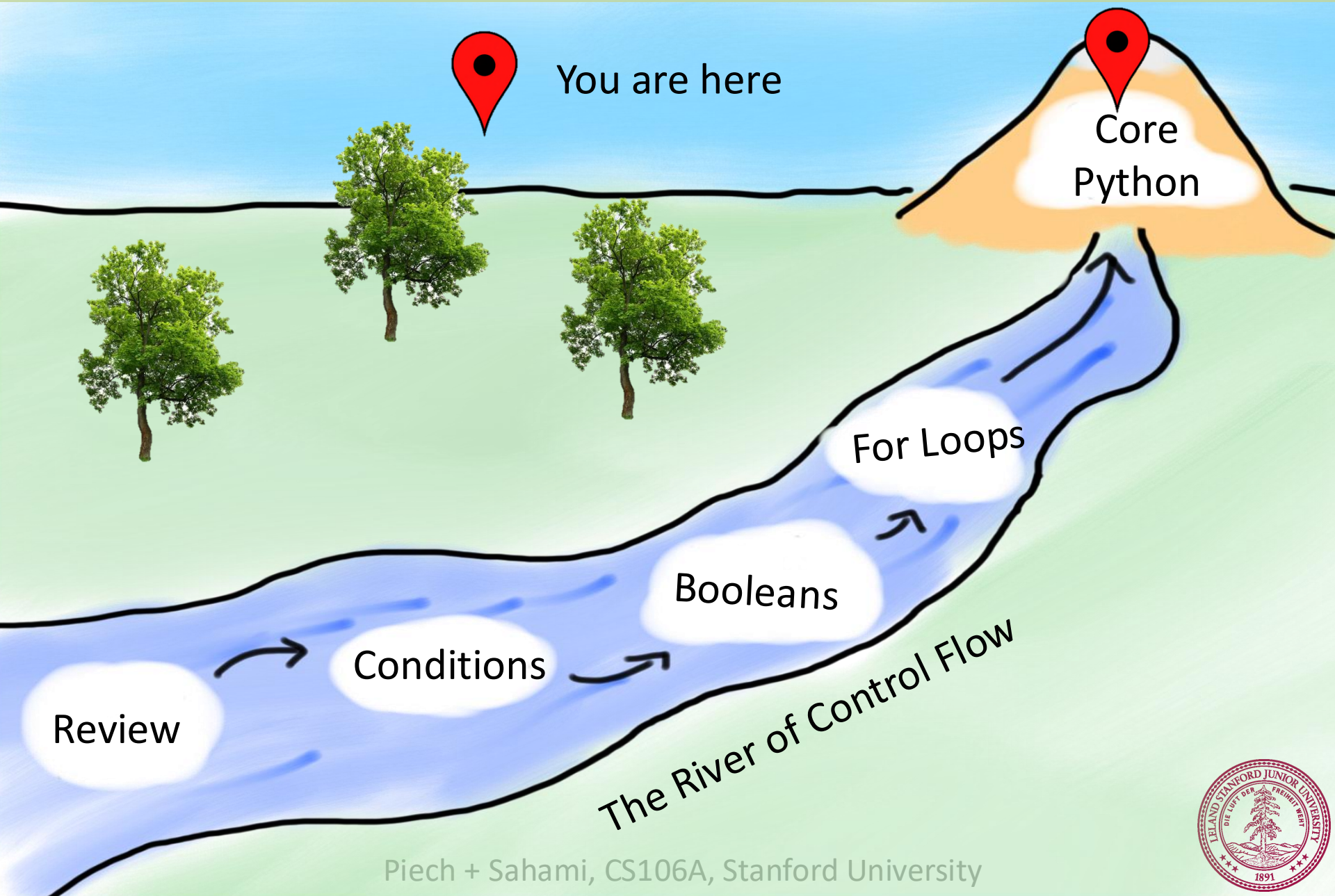


That's all

```
def main() :
    for i in range(999999):
        print("You rock!")
        print("Keep learning!")
```




Today's Route



Today's Goal

1. Be able to use For / While / If in Python

